



histopathology Inflammation part II

3rd class Lecture 4

Assistant Prof. Dr. Rasha A. AL Safi

histopathologist

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OBJECTIVES:

1. List four morphologic patterns of acute inflammation and describe at least one example of each.
2. The beneficial and harmful effects of acute inflammation, and summarize the three possible outcomes following an inflammatory episode.
3. Identify key cellular features that distinguish acute from chronic inflammation, such as the presence of mononuclear infiltrates and fibrosis.
4. Analyze the roles of macrophages, lymphocytes, plasma cells, and eosinophils in chronic inflammation and describe how their interactions sustain the inflammatory process.
5. Evaluate the clinical relevance of chronic inflammation in diseases such as tuberculosis, atherosclerosis, and rheumatoid arthritis based on pathologic findings.
6. Construct a comparative table summarizing acute versus chronic inflammation including causes, cells involved, morphological features, and outcomes.
7. recall at least five diseases associated with granulomatous inflammation and define the term “granuloma” with its cellular composition.
8. explain the differences between foreign body and immune granulomas and describe the role of cytokines like IFN- γ and IL-2 in their formation.
9. identify granulomas, differentiate caseating from non-caseating forms, and relate findings to diseases like TB or sarcoidosis.

MORPHOLOGIC PATTERNS OF ACUTE INFLAMMATION

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I. Serous inflammation:

- Is the outpouring of a **watery**, relatively **protein-poor fluid** , depending on the site of injury.

- Typically, the fluid :
 1. is **not infected**.
 2. does **not** contain large numbers of leukocytes.

- In **body cavities** the fluid derived from:
 - the **plasma** (as a result of increased vascular permeability) or
 - from the **mesothelial cells** (as a result of local irritation).

- Accumulation of fluid in **body cavities** is called an **effusion**.
- Effusions in non-inflammatory conditions consisting of:
 1. Reduced blood outflow in HF.
 2. Reduced plasma protein levels in kidney and liver diseases.

E,g

- The **skin blister** resulting from a burn or viral infection:
 - a serous effusion either **within** or immediately beneath the **epidermis** of the skin .

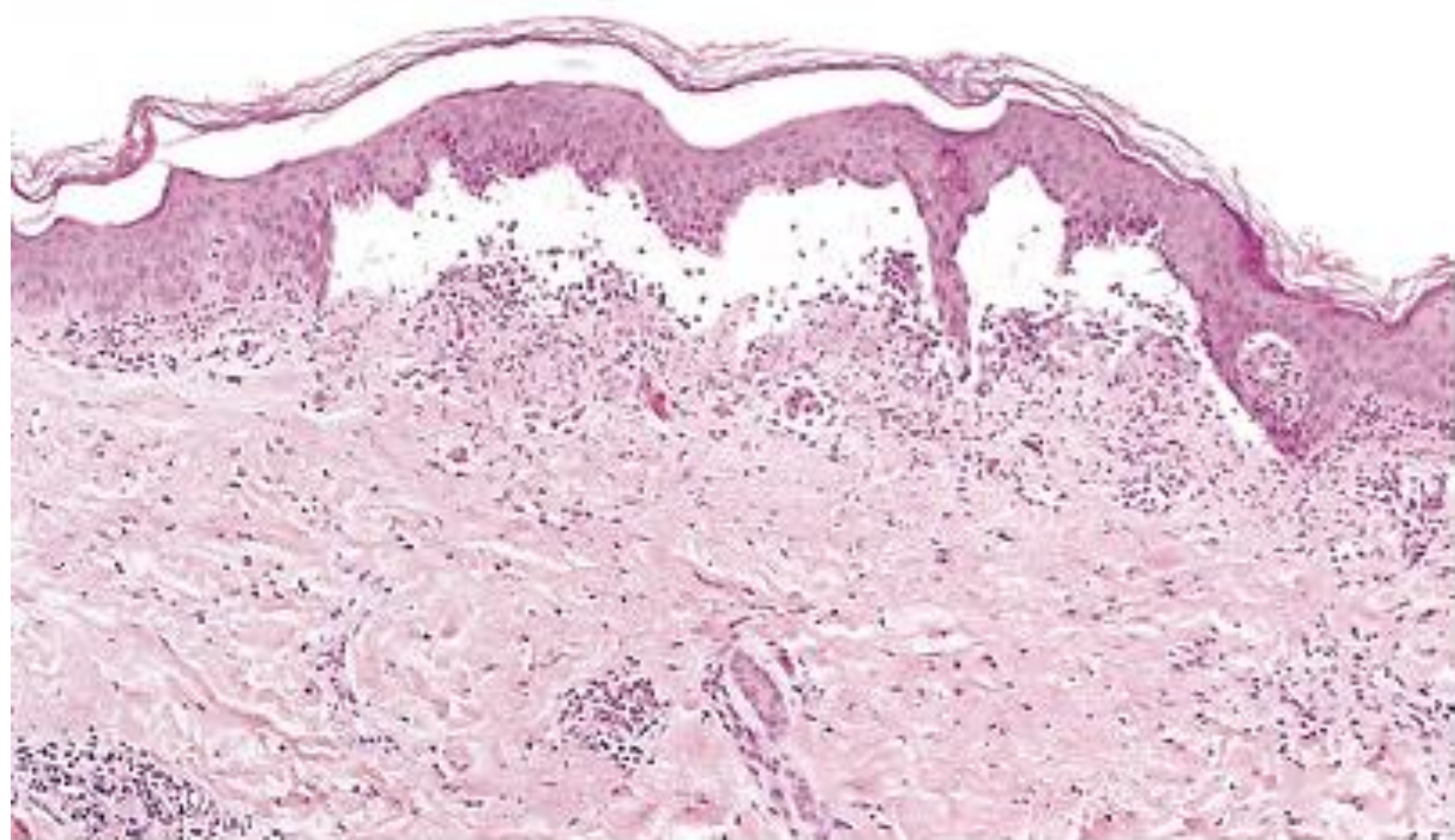


Fig. 3.12 Serous inflammation. Low-power view of a cross section of a skin blister showing the epidermis separated from the dermis by a focal collection of serous effusion.

MORPHOLOGIC PATTERNS OF ACUTE INFLAMMATION

II. Fibrinous inflammation:

- occurs as a consequence of more severe injuries, resulting in greater vascular permeability that allows large molecules (such as **fibrinogen**) to pass the endothelial barrier.
- Such exudates may be ended with:

1. Resolution

- The exudate degraded by fibrinolysis.
- Accumulated debris may be removed by macrophages.
- Restoration of the normal tissue structure.

2. Organization & scarring:

- Extensive fibrin-rich exudates may not be completely removed.
 - Exudate replaced by an ingrowth of fibroblasts and blood vessels.
 - **Scarring.**
- A fibrinous exudate is characteristic of inflammation in the lining of body cavities, such as the **meninges, pericardium.**

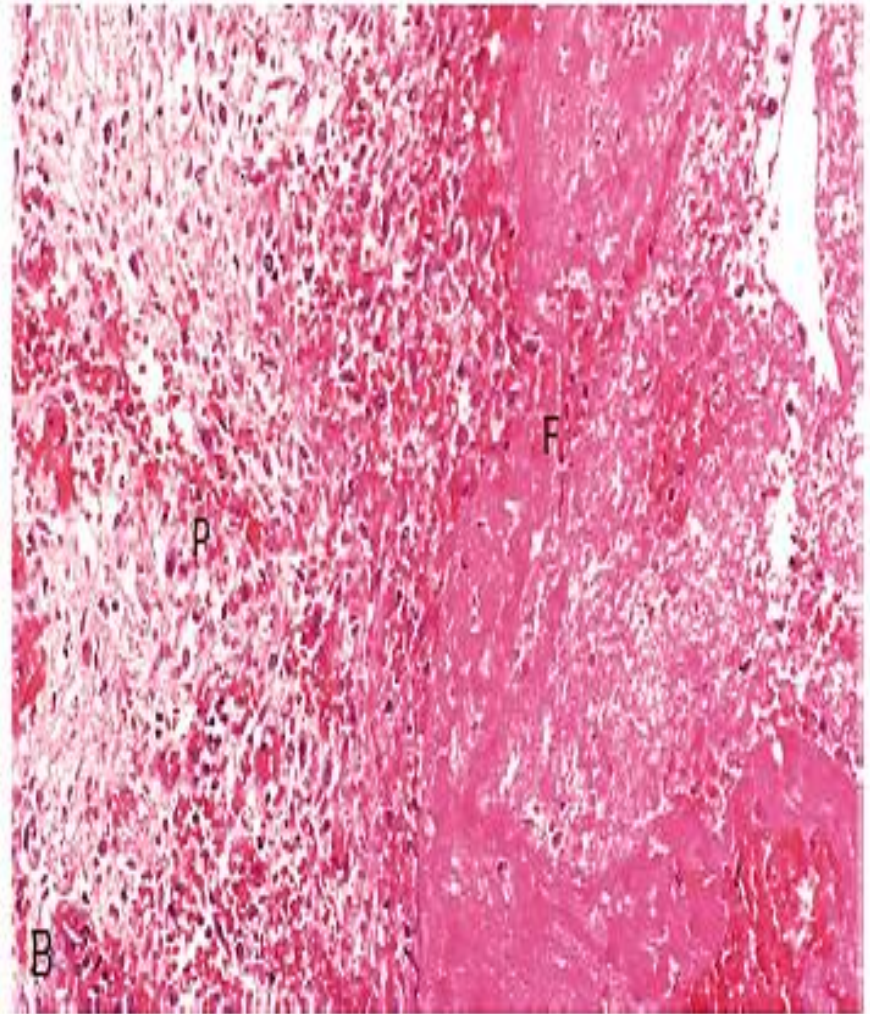


Figure 2-12 Fibrinous pericarditis. **A**, Deposits of fibrin on the pericardium. **B**, A pink meshwork of fibrin exudate (F) overlies the pericardial surface (P).

MORPHOLOGIC PATTERNS OF ACUTE INFLAMMATION

III. Suppurative (purulent) inflammation & abscess formation.

- These are manifested by the **collection** of large amounts of **purulent exudate** (pus) consisting of neutrophils, necrotic cells, and edema fluid.
- Certain organisms (e.g., **staphylococci**) are more likely to induce such localized suppuration and are therefore referred to as **pyogenic** (pus-forming).
- **Abscesses: are localized collections of pus** caused by suppuration buried in a tissue, an organ, or a confined space. They are produced by seeding of pyogenic bacteria into a tissue.

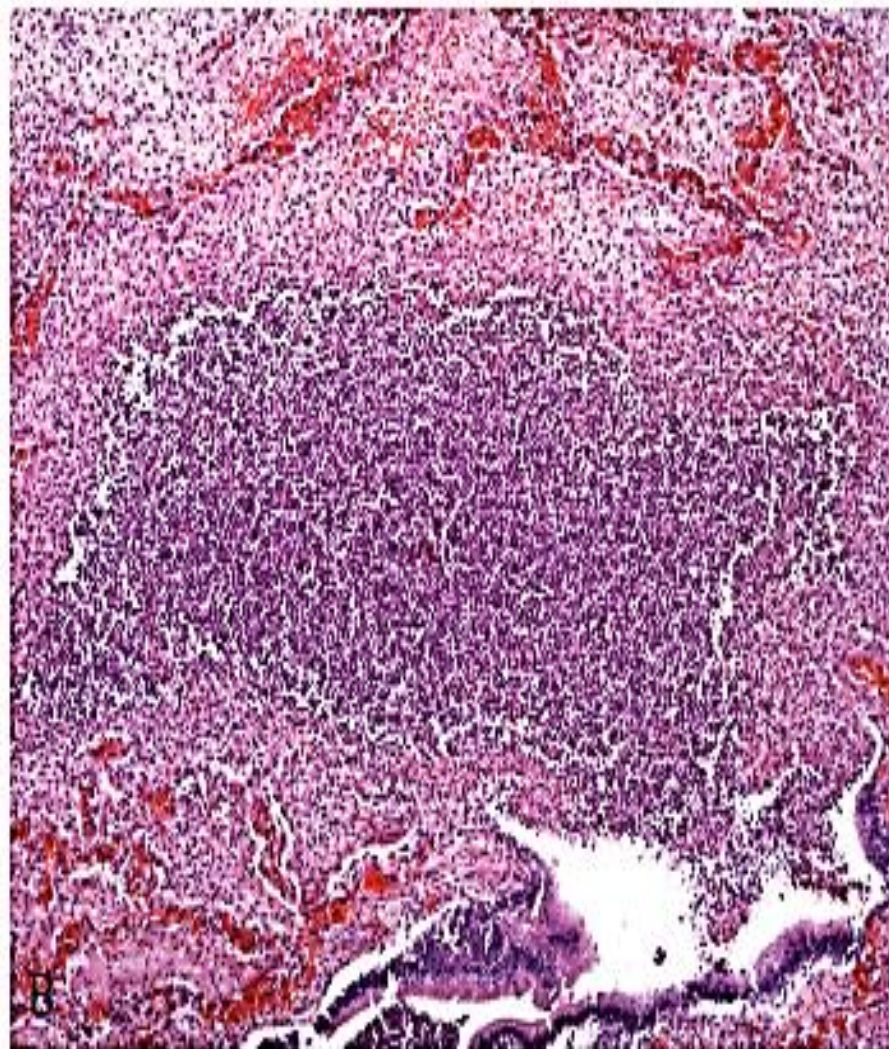


Figure 2-13 Purulent inflammation with abscess formation. **A**, Multiple bacterial abscesses in the lung (*arrows*) in a case of bronchopneumonia. **B**, The abscess contains neutrophils and cellular debris and is surrounded by congested blood vessels.

MORPHOLOGIC PATTERNS OF ACUTE INFLAMMATION

An ulcer:

- A local **defect**, or **excavation**, of the surface of an organ or tissue that is produced by **necrosis** of cells and **sloughing** (shedding) of necrotic and inflammatory tissue.
 - Ulceration can occur only when **tissue necrosis** & resultant **inflammation** exist on or near a surface.
 - It is most commonly encountered in:
 1. The **mucosa** of the mouth, stomach, intestines, or genitourinary tract, and
 2. The **skin** and subcutaneous tissue of the lower extremities in older persons with extensive ischemic necrosis.

- E.g:
- **Peptic ulcer** of the stomach or duodenum.
- Acute and chronic inflammation coexist.

- During the **acute stage**:
 - Intense PMN infiltration and
 - Vascular dilation in the margins of the defect.

- With **chronicity**:
 - the margins and base of the ulcer develop **scarring**.
 - accumulation of lymphocytes, macrophages, and plasma cells.

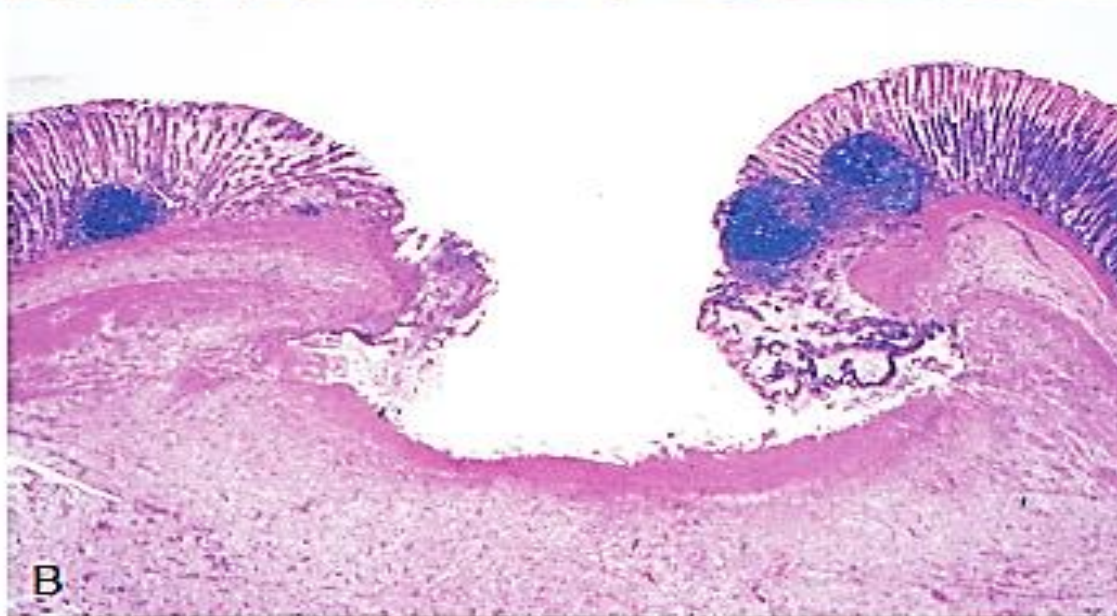


Figure 2-14 Ulcer. A, A chronic duodenal ulcer. **B,** Low-power cross-section of a duodenal ulcer crater with an acute inflammatory exudate in the base.

EFFECTS OF ACUTE INFLAMMATION

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Beneficial Effects

1. Dilution of Toxins by the edema fluid.
2. Production of protective Antibodies & promotion of immunity.
3. Fibrin meshwork formation that forms a scaffold for inflammatory cell migration & also limits the spread of infections.
4. Cell Nutrition.

Harmful Effects

1. Swelling & edema that can be detrimental for e.g. acute epiglottitis that may be life threatening.
2. Rise in tissue pressure that contributes to tissue necrosis.
3. Digestion of adjacent viable tissue.
4. Sever damaging allergic reaction.
5. Generalized increase in vascular permeability can cause shock as seen in anaphylactic reactions.

Laryngeal edema due to acute epiglottitis



Note congestion and the marked narrowing of laryngeal orifice

OUTCOMES OF ACUTE INFLAMMATION

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In general, acute inflammation may have one of three outcomes

1. Complete resolution: when

- a. the injury is limited or short-lived.
- b. there has been little tissue destruction.
- c. the damaged parenchymal cells can regenerate.

2. Healing by fibrosis

- a. after extensive tissue destruction.
- b. when the inflammatory injury involves tissues that are incapable of regeneration.
- c. when there is abundant fibrin exudation.

3. Progression to chronic inflammation

- Acute to chronic transition occurs when the acute inflammatory response persists, owing either to the perseverance of the injurious agent or to some interference with the normal process of healing.

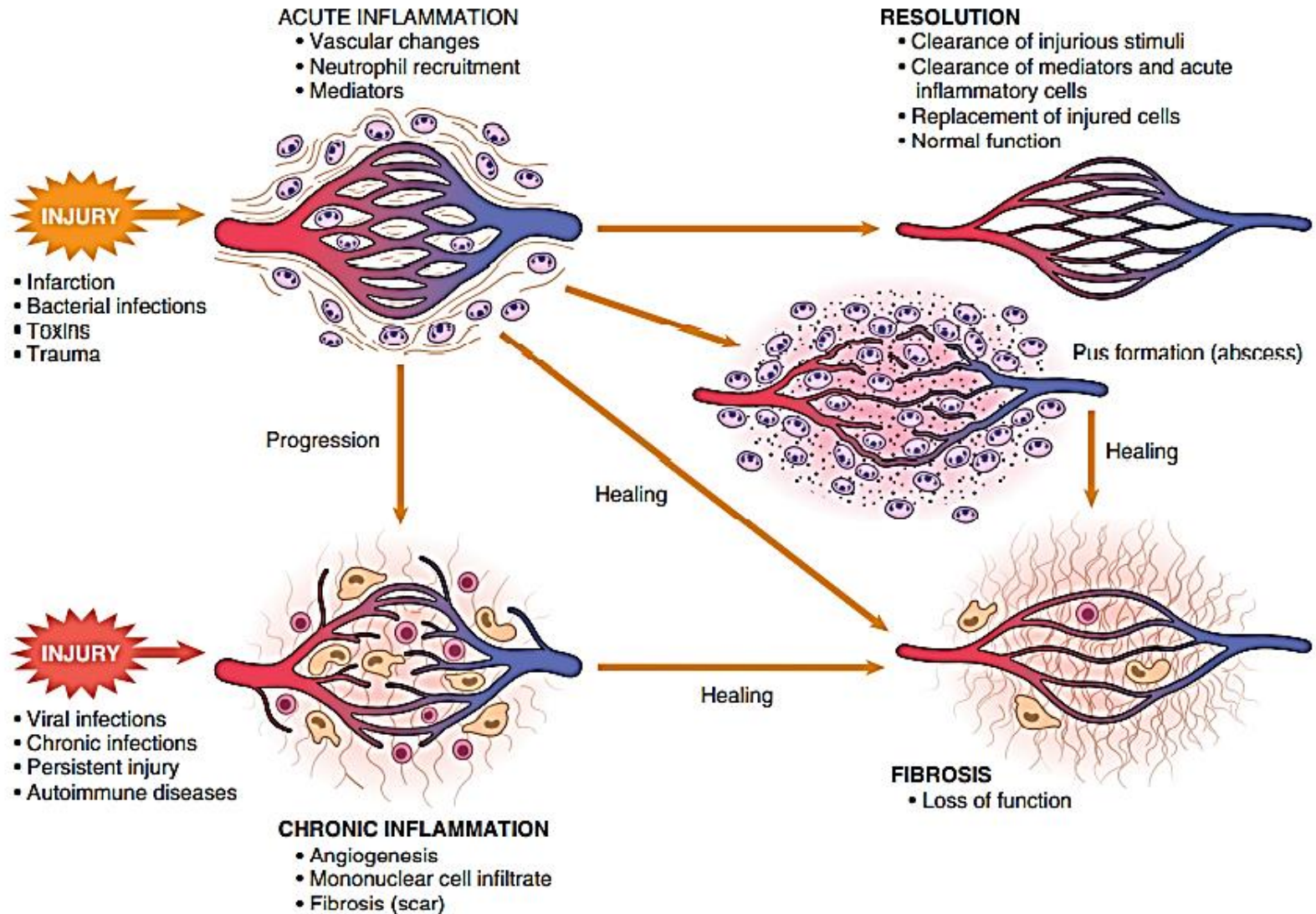


Fig. 3.16 Outcomes of acute inflammation: resolution, healing by fibrosis, or chronic inflammation. The components of the various reactions and their functional outcomes are listed.

CHRONIC INFLAMMATION

CHRONIC INFLAMMATION

- Chronic inflammation is the cause of tissue damage in some of the most common and disabling human diseases, such as:
 1. Rheumatoid arthritis.
 2. Atherosclerosis.
 3. Tuberculosis.
 4. Chronic lung diseases.
- It frequently **begins** from the outset **as a chronic** (chronic inflammation ab initio), **insidious**, and **low-grade** response.
- Chronic Inflammation may complicate acute inflammation which is almost always a **suppurative type** of inflammation that presents as a purulent discharge (**pus**) as seen in **abscess**.

Causes of chronic inflammation ab initio include

1. Persistent infections:

- by certain microorganisms such as *tubercle bacilli*, *Treponema pallidum*, certain *viruses*, *fungi*, and *parasites*.
 - These organisms are of **low toxicity** and
 - evoke **delayed hypersensitivity** reaction.
 - The inflammatory response sometimes takes a specific pattern called **granulomatous** inflammation.

2. Prolonged exposure to toxic agents:

- **Exogenous** : as inhaled silica particles, or
- **Endogenous**: such as toxic plasma lipids that are responsible for atherosclerosis.
 - atherosclerosis is thought to be a chronic inflammatory process of the arterial wall.

3. Hypersensitivity diseases.

Morphologic features of chronic inflammation

1. Infiltration with mononuclear cells:

- Macrophages .
- Lymphocytes.
- plasma cells.

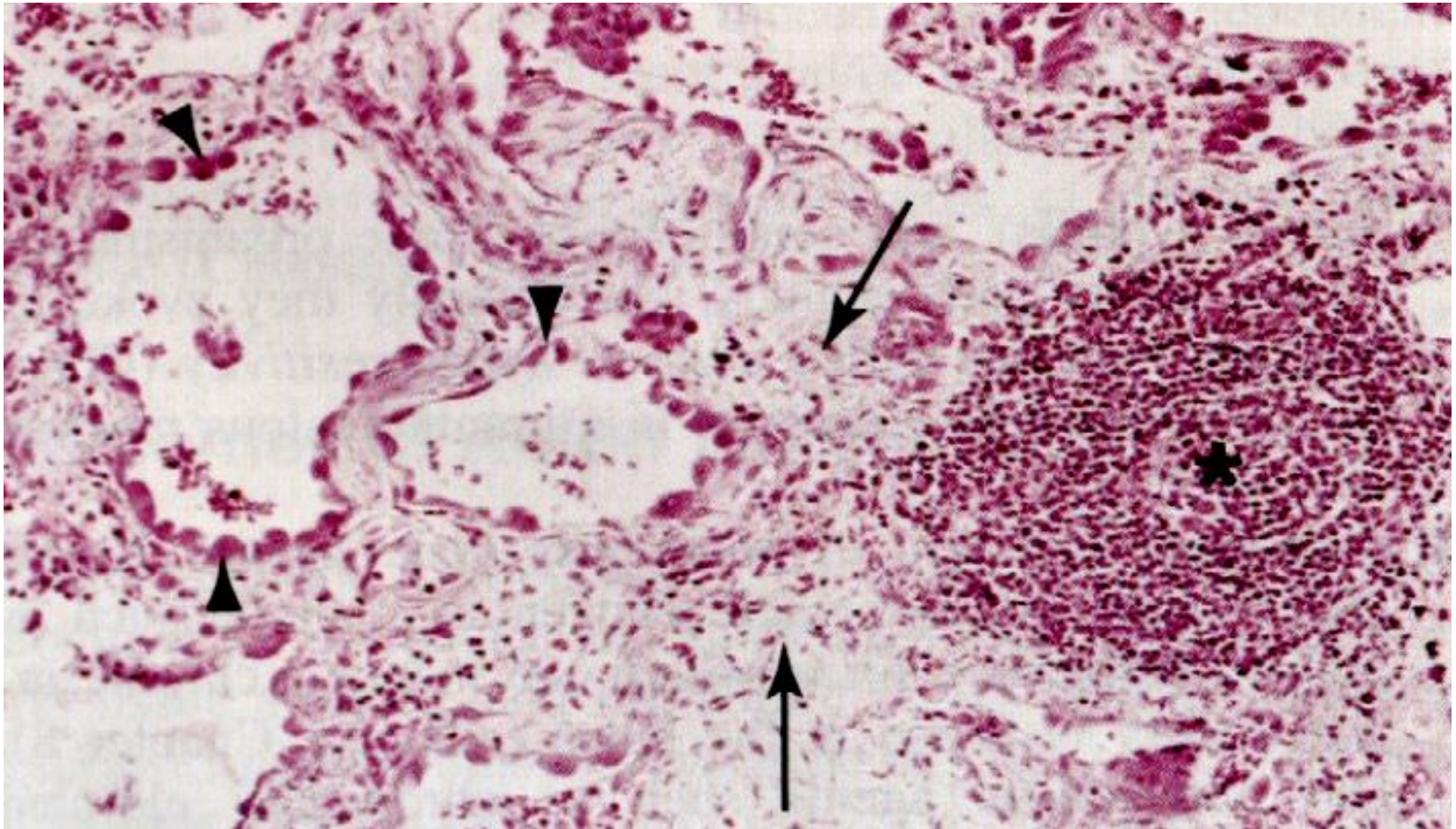
2. Tissue destruction: induced by

- Persistent offending agent or
- The inflammatory cells.

3. Attempts at healing by fibrosis of the damaged tissue.

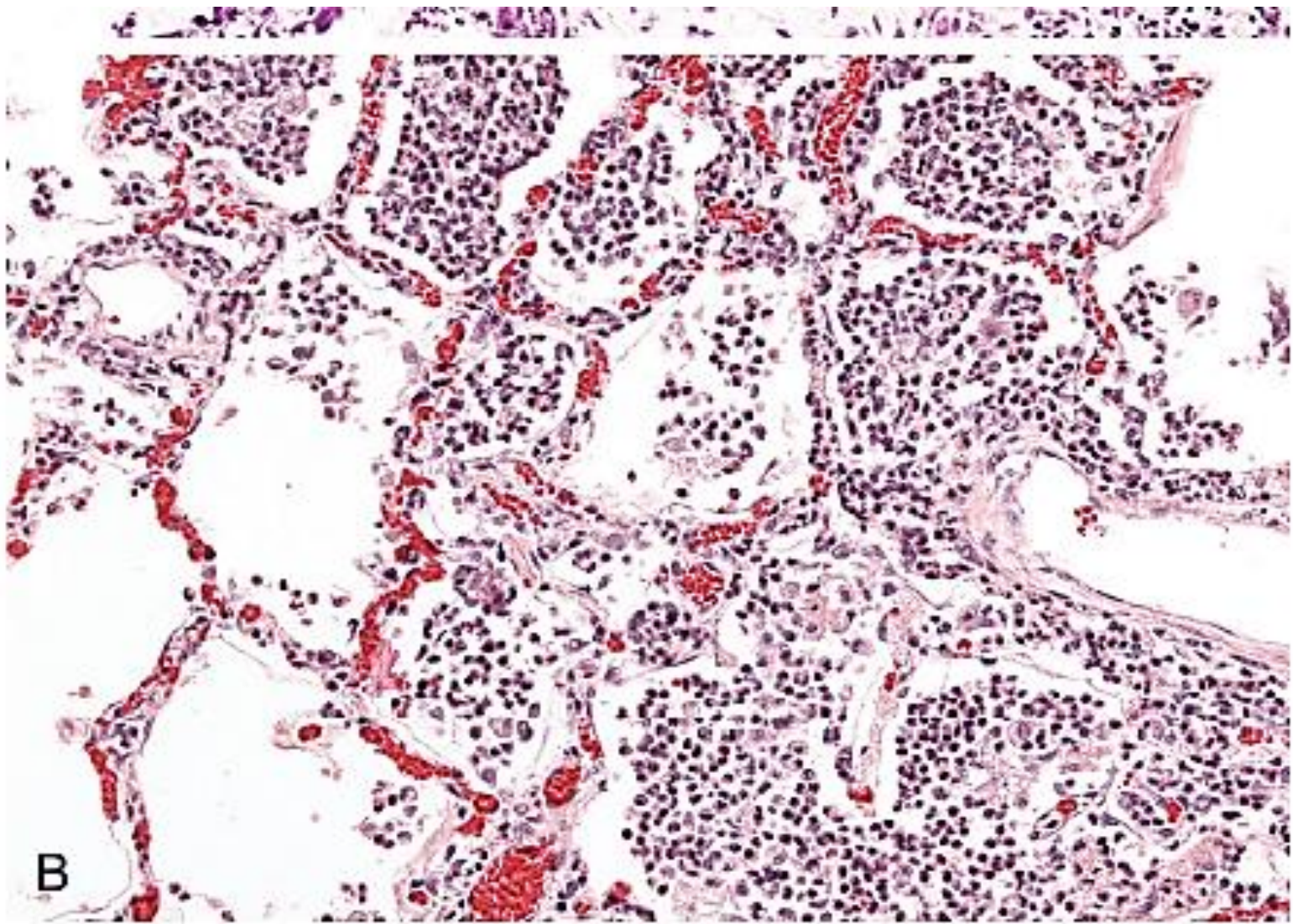
- achieved by proliferation of **small blood vessels (angiogenesis) & fibroblasts.**

Features of chronic inflammation



The three characteristic features of chronic inflammation (in the lung)

- 1. Chronic inflammatory cells infiltration***
- 2. destruction of the normal tissue (normal alveoli are replaced by spaces lined by cuboidal cells (arrow heads))**
- 3. Replacement by fibrosis (arrows)**



acute inflammation of the lung (acute bronchopneumonia), neutrophils fill the alveolar spaces and blood vessels are congested.

Mononuclear cell infiltration

I. The macrophage:

- The dominant cells in chronic inflammation.
- It is diffusely scattered in connective tissues or located **in organs**:
 - **in the liver** (Kupffer cells).
 - In **spleen** and **lymph nodes** (sinus histiocytes).
 - In **lungs** (alveolar macrophages).

From the blood, monocytes **migrate** into various tissues and differentiate into macrophages.

- The half-life of blood monocytes is about 1 day.
- The life span of tissue macrophages is several months or years.
- There are two major pathways of macrophage activation, called **classical** and **alternative pathways**.

– In chronic inflammation, **macrophage accumulation persists**, and this is mediated by the following:

1. **Recruitment** from circulating monocytes.
2. **Local proliferation** of macrophages after their emigration from the bloodstream.
3. **Immobilization** of macrophages within the site of inflammation.

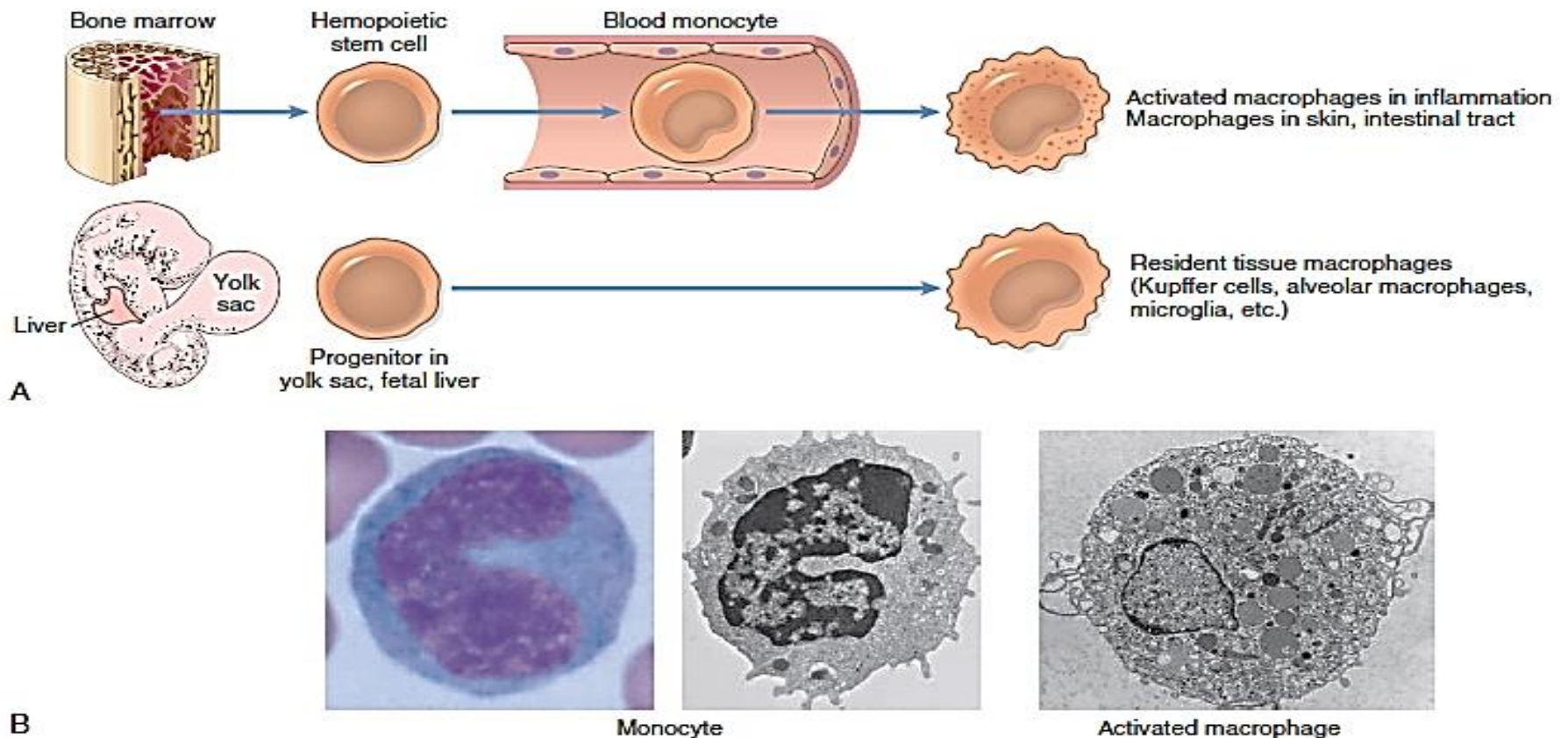


Fig. 3.18 Maturation of mononuclear phagocytes.

(A) During inflammatory reactions, the majority of tissue macrophages are derived from hematopoietic precursors. Some long-lived resident tissue macrophages are derived from embryonic precursors that populate the tissues early in development.

(B) The morphology of a monocyte and activated macrophage.

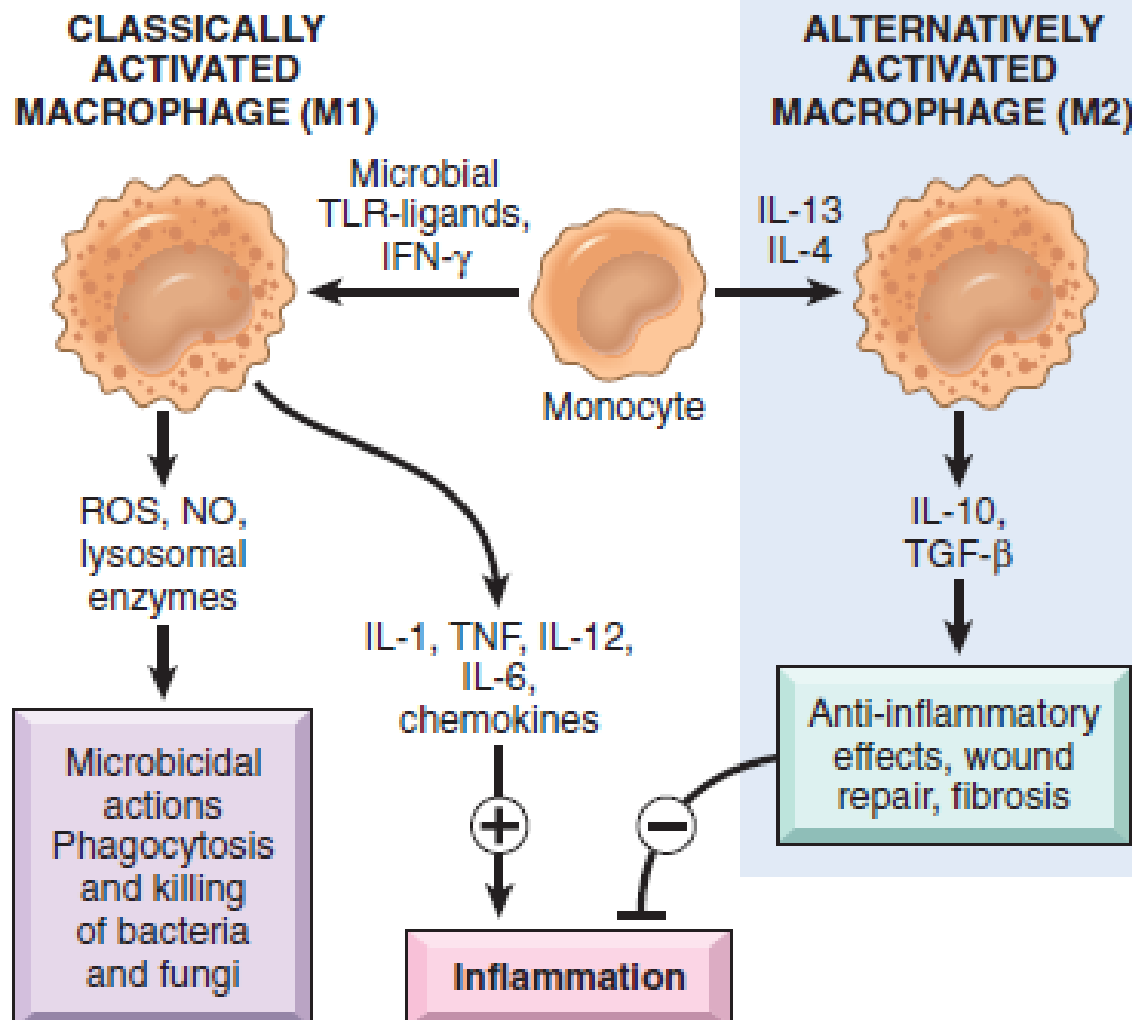


Fig. 3.19 Classical and alternative macrophage activation. Different stimuli activate monocytes/macrophages to develop into functionally distinct populations. Classically activated macrophages are induced by microbial products and cytokines, particularly IFN- γ . They phagocytose and destroy microbes and dead tissues and can potentiate inflammatory reactions. Alternatively activated macrophages are induced by other cytokines and are important in tissue repair and the resolution of inflammation.

The products of activated macrophages:

- are responsible for much of the tissue injury in chronic inflammation; these **products include**
 1. **Toxic** substances to microbes and host cells (e.g., toxic O₂ species, NO, and proteases).
 2. Chemo-attractants to other inflammatory cells.
 3. Growth factors: the cause of fibroblast proliferation, collagen deposition, and angiogenesis.

Functions:

- Macrophages **display antigens** to **T lymphocytes** and respond **to signals** from **T cells**, thus setting up a feedback loop.
- Macrophages secrete mediators of inflammation, like cytokines (**TNF, IL-1, chemokines**)
- The **products** of activated macrophages eliminate injurious agents and initiate the process of **repair**.

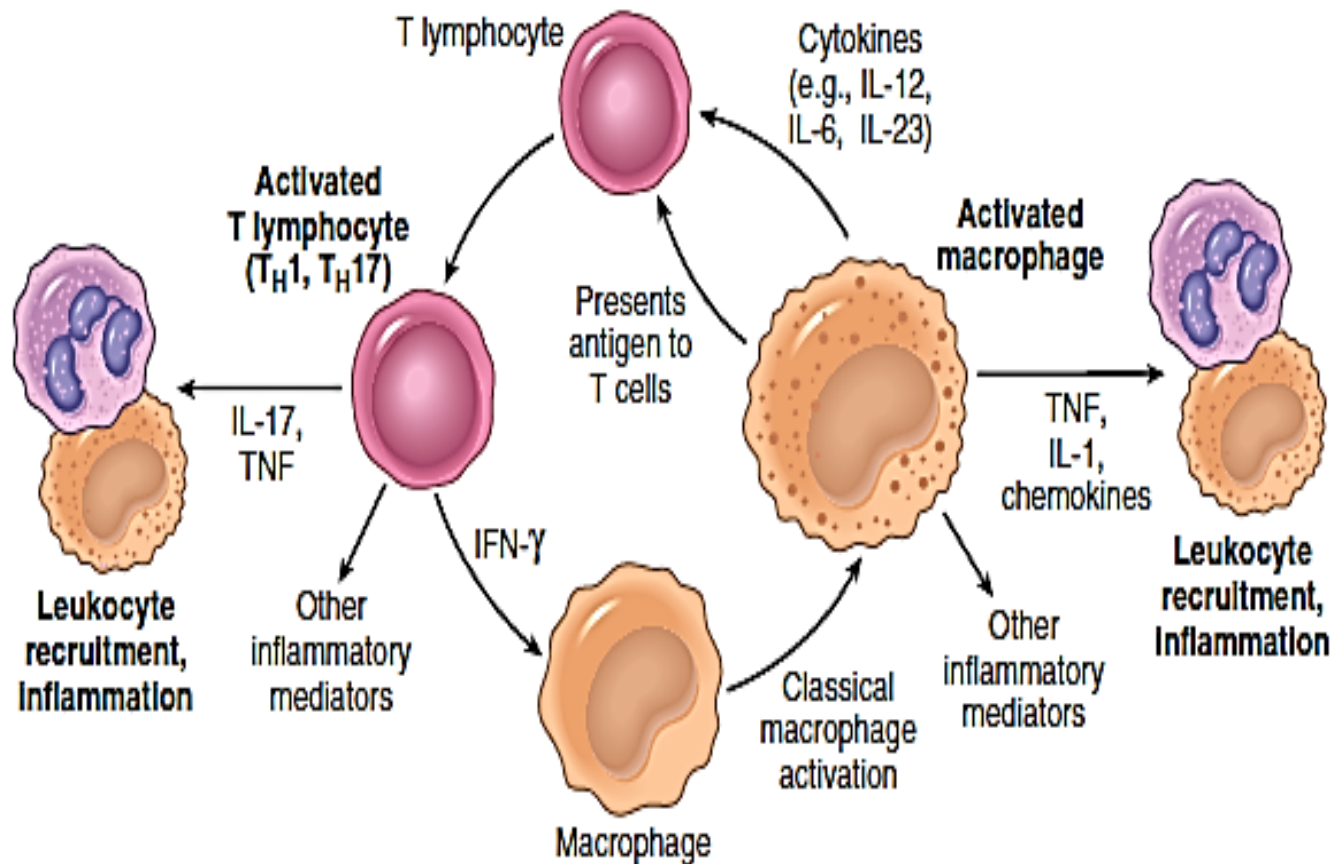


Fig. 3.20 Macrophage–lymphocyte interactions in chronic inflammation. Activated T cells produce cytokines that recruit macrophages (TNF, IL-17, chemokines) and others that activate macrophages (IFN- γ). Activated macrophages in turn stimulate T cells by presenting antigens and via cytokines such as IL-12.

- Lymphocytes and macrophages interact in a bidirectional way, and these interactions play an important role in propagating chronic inflammation.
- The result is a cycle of cellular reactions that fuel and sustain chronic inflammation.

II. Lymphocytes

- Lymphocytes and macrophages interact in a bidirectional way.
- These reactions play an important role in chronic inflammation.
- Macrophages display antigens to T cells that stimulate them.
- **Activated T lymphocytes** produce cytokines, **IFN- γ** , which is a major activator of macrophages.
- **CD4+ T lymphocytes** promote inflammation by secreting **cytokines**.

- There are **three subsets of CD4+ T** cells that secrete different cytokines and elicit different types of inflammation.

1. **TH1 cells:**

- Produce the cytokine **IFN- γ**
- **IFN- γ** activates macrophages by the classical pathway.

2. **TH2 cells:**

- secrete IL-4, IL-5, and IL-13.
- These recruit and activate eosinophils, against parasites and in allergic inflammation.
- are responsible for the alternative pathway of macrophage activation.

3. **TH17 cells:**

- secrete **IL-17** and other cytokines.
- These induce the secretion of **chemokines** responsible for recruiting neutrophils into the reaction.

III. Plasma cells:

- develop from activated B lymphocytes and
- produce antibody directed against persistent antigen in the inflammatory site.

IV. Eosinophils:

- are abundant in **immune reactions** mediated by **IgE** .
- in parasitic infections.
- One of the **chemokines** that is especially important for eosinophil recruitment is **eotaxin**.
- Eosinophils have granules that contain major basic protein that is toxic to parasites.

V. Mast cells:

- are widely distributed in connective tissues.
- participate in both acute and persistent inflammatory reactions.
- Mast cells express on their surface the receptor that binds the IgE antibody.
- Mast cells are also present in chronic inflammatory reactions, and may produce cytokines that contribute to fibrosis.

VI. Neutrophils:

- Many forms of chronic inflammation show large numbers of neutrophils.
- Presence of neutrophils induced either by:
 1. Persistent microbes.
 2. Mediators produced by macrophages and T lymphocytes.
- E.g:
 1. In chronic bacterial infection of bone (**osteomyelitis**), a neutrophilic exudate can persist for many months.
 2. Neutrophils are also important in the chronic damage induced in lungs by smoking and other irritant stimuli.

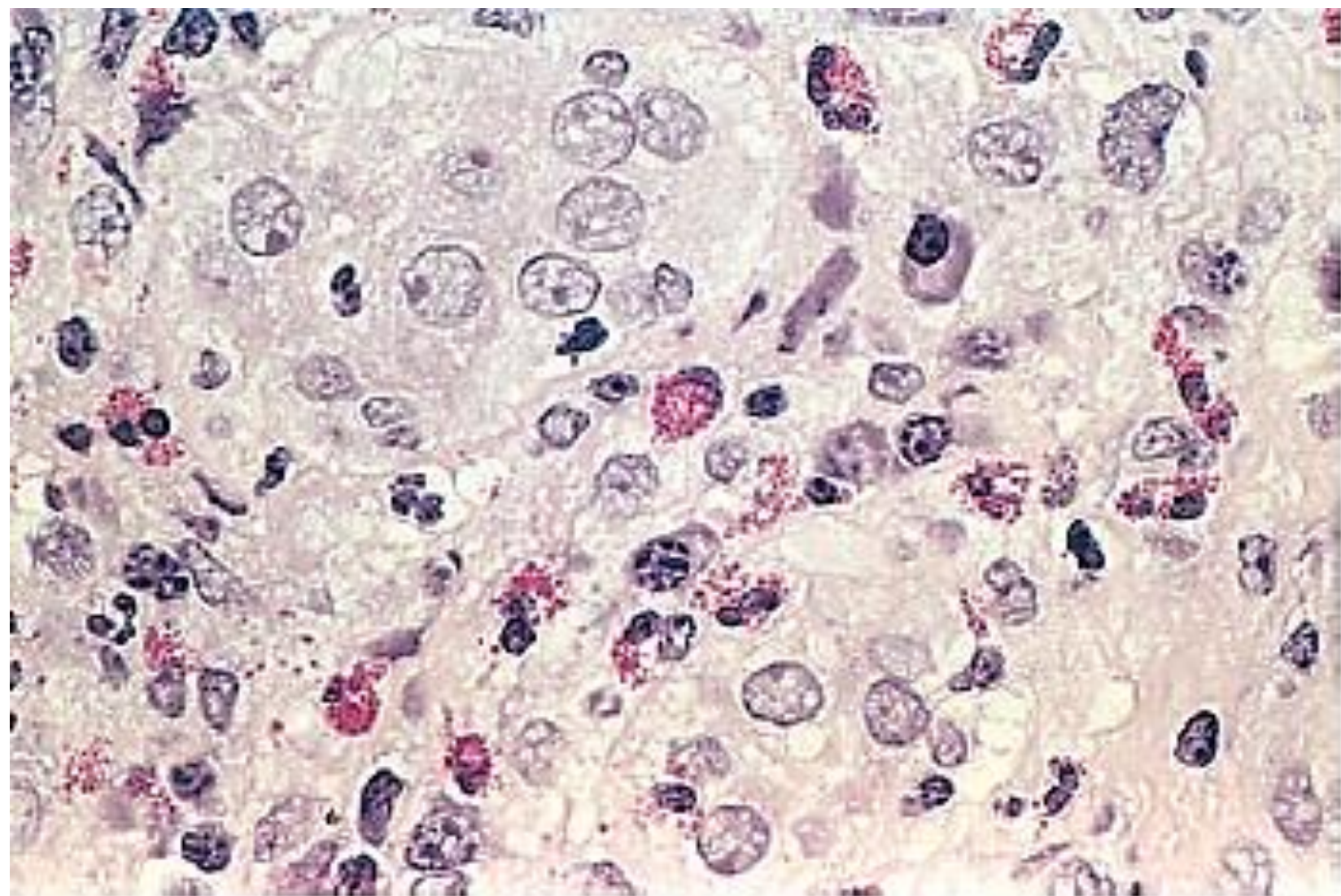


Fig. 3.21 A focus of inflammation containing numerous eosinophils.

Chronic cholecystitis with cholelithiasis



Note thickening of the wall due to fibrosis

GRANULOMATOUS INFLAMMATION

- This is a distinctive pattern of chronic inflammatory reaction characterized by focal accumulations of activated macrophages, which often develop an epithelioid (epithelial-like) appearance.

Causes

- Granulomatous inflammation is encountered in a number of immunologically mediated infectious and some noninfectious conditions, these include:
 1. Tuberculosis.
 2. Sarcoidosis
 3. Cat-scratch disease
 4. Lymphogranuloma inguinale.
 5. Leprosy.
 6. Brucellosis
 7. Syphilis.
 8. fungal infections.
 9. Berylliosis
 10. Reactions of irritant lipids.

Granuloma

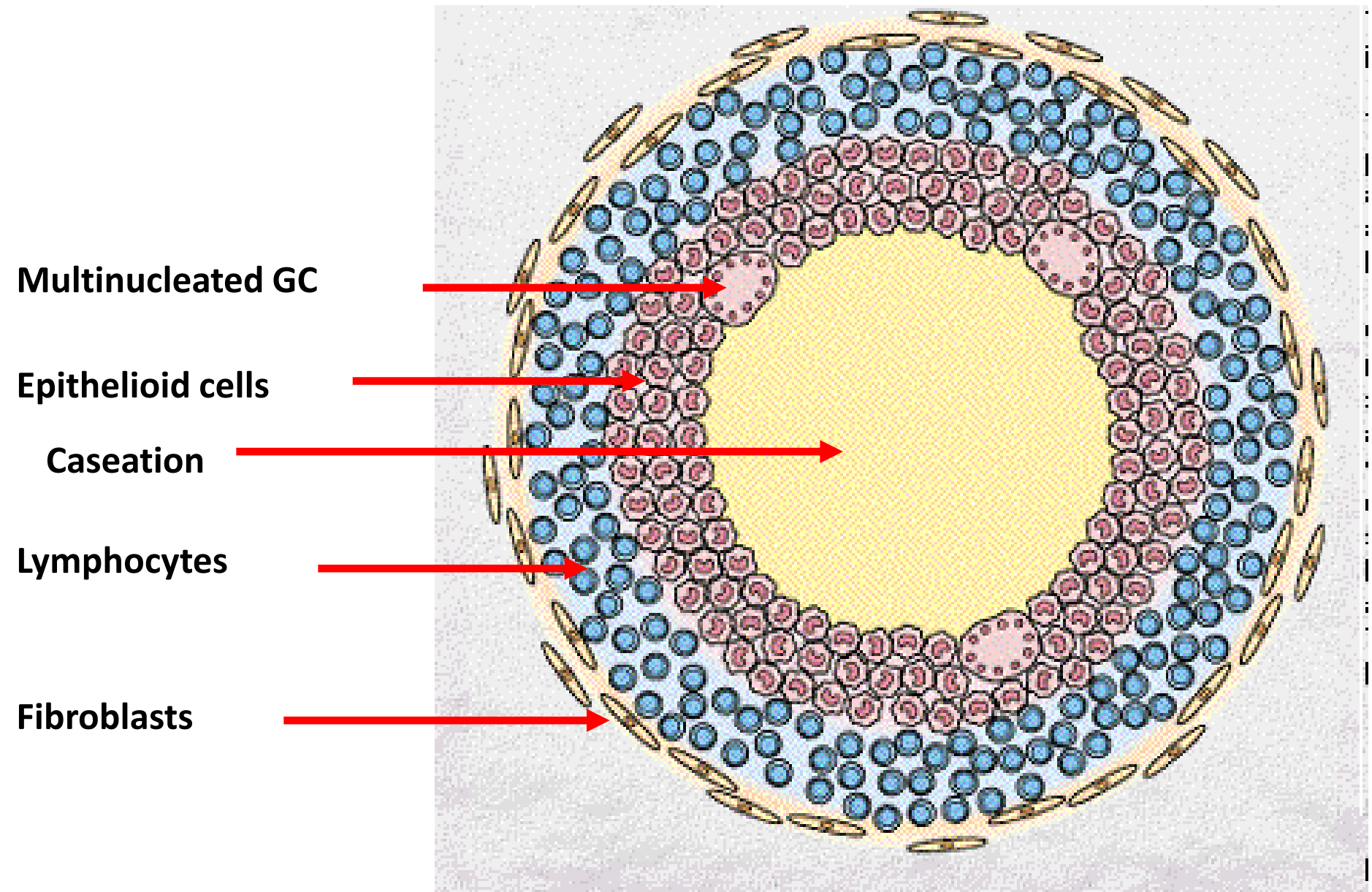
- A granuloma is a focus of chronic inflammation consisting of a microscopic aggregation of macrophages that are transformed into epithelioid cells surrounded by a collar of mononuclear leukocytes, principally lymphocytes and occasionally plasma cells.
- The epithelioid cells have a pale pink granular cytoplasm with indistinct cell borders and a vesicular nucleus that is oval or elongate.
- Older granulomas develop an enclosing rim of fibroblasts and connective tissue.

- Frequently, epithelioid cells fuse to form giant cells in the periphery or in the center of granulomas.
- These **giant cells** may attain diameters of 40 to 50 μm .
- They have a large mass of cytoplasm containing 20 or more small nuclei arranged either peripherally (Langhans-type giant cell) or haphazardly (foreign body-type giant cell).

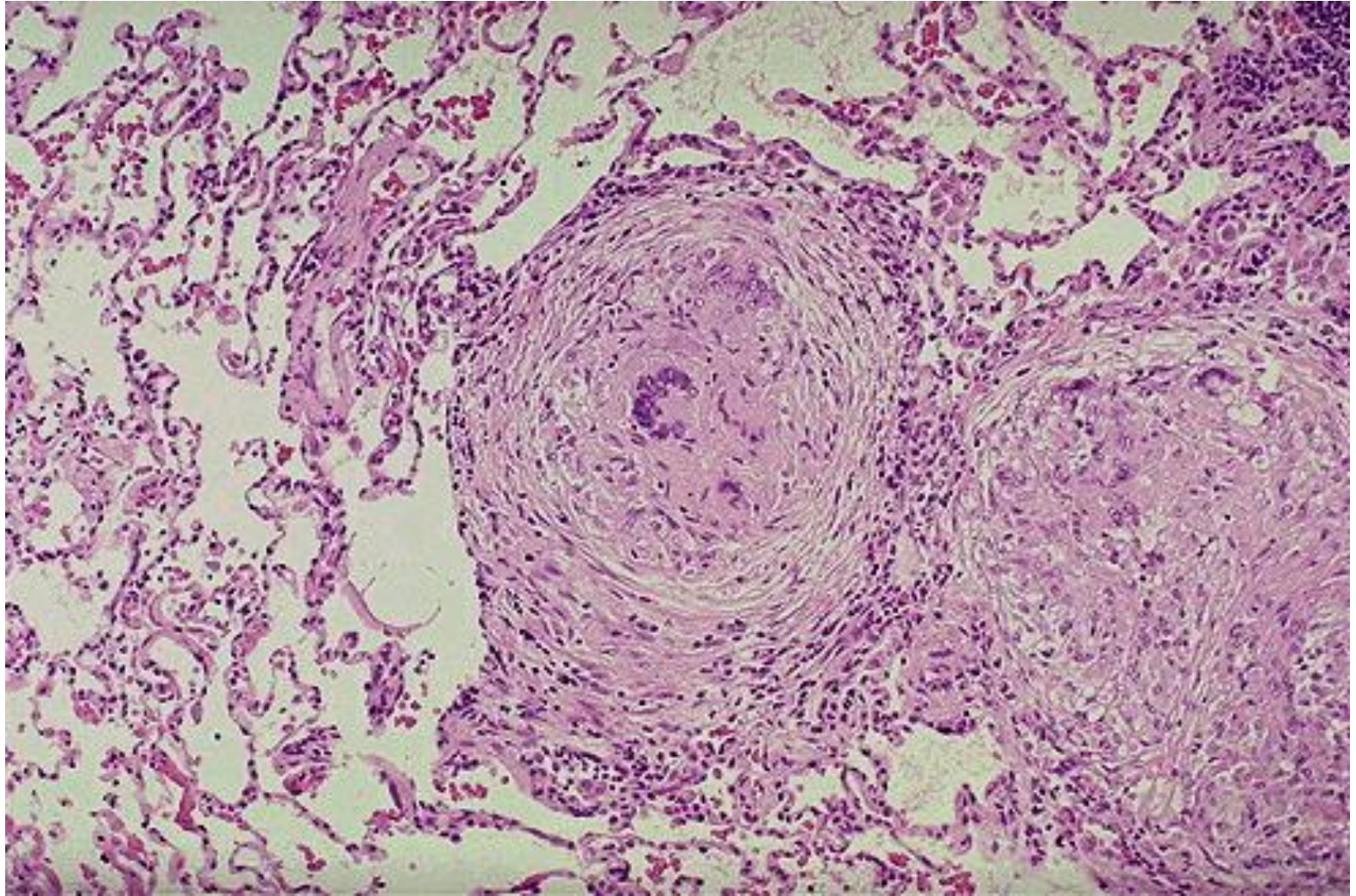
Granuloma

- Granulomas associated with certain infectious organisms (Mycobacterium tuberculosis) often contain a central zone of necrosis.
 - Grossly, this has a granular, cheesy appearance and is therefore called caseous necrosis.
 - Microscopically, this necrotic material appears as amorphous, structureless, eosinophilic, granular debris, with loss of cellular details.
- The granulomas in Crohn disease, sarcoidosis, and foreign body reactions tend to not have necrotic centers and are non-caseating.
- Healing of granulomas is accompanied by fibrosis that may be extensive.

Diagram of typical TB granuloma

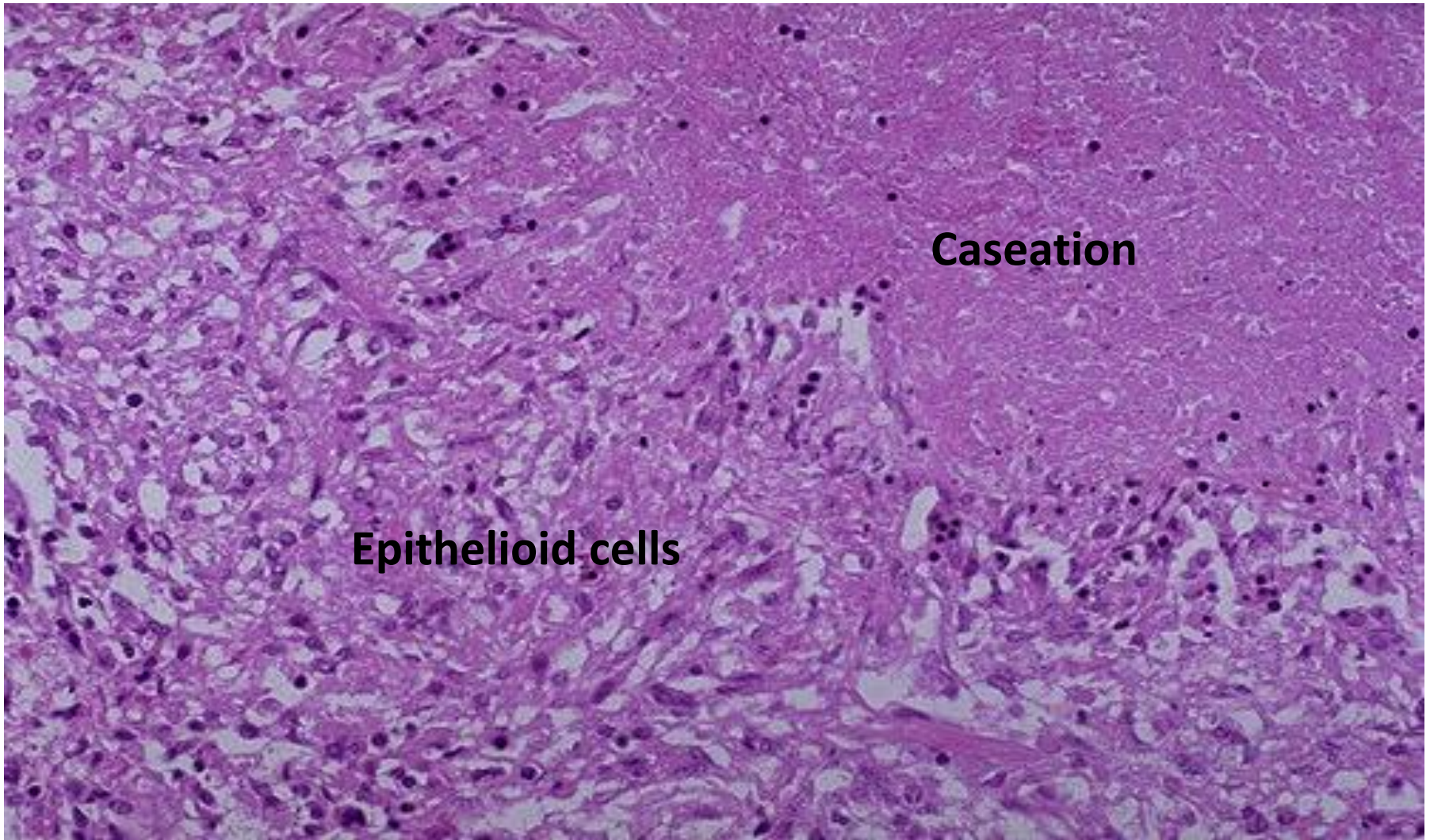


TB granulomas lung



This is a low power view showing two, adjacent, well-defined, rounded granulomas . From this power the presence of multinucleated giant cells is obvious.

TB granulomas



Epithelioid cells

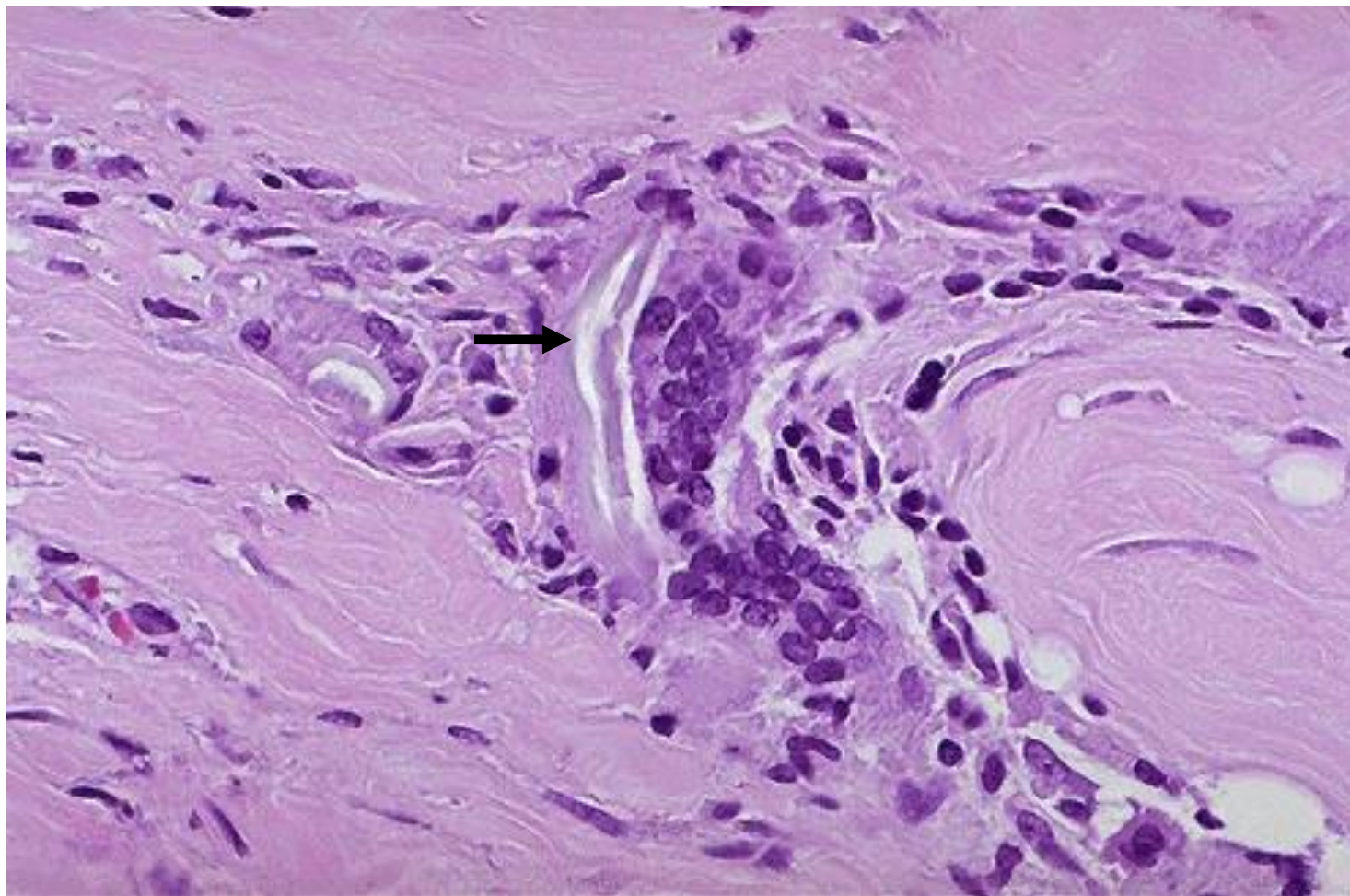
Caseation

This is a high power view showing a portion of typical TB granuloma. Note the amorphous, pinkish central caseation, which is surrounded by a rim of epithelioid cells.

1. Foreign body granulomas

- Foreign body granulomas form when material that can not be phagocytosis by a single macrophage such as:
 1. Talc.
 2. Sutures.
 3. fibers
- **Epithelioid** cells and **giant** cells form and are **apposed** to the surface of the foreign body and/or actually include it.
- The foreign material can usually be identified in the center of the granuloma.
- Viewed with polarized light as a refractile.

Foreign body giant cells in suture granuloma



Two foreign body giant cells are seen, where there is a bluish strand of suture material (arrow) from a previous operation

Talc granulomas, pulmonary, polarized light



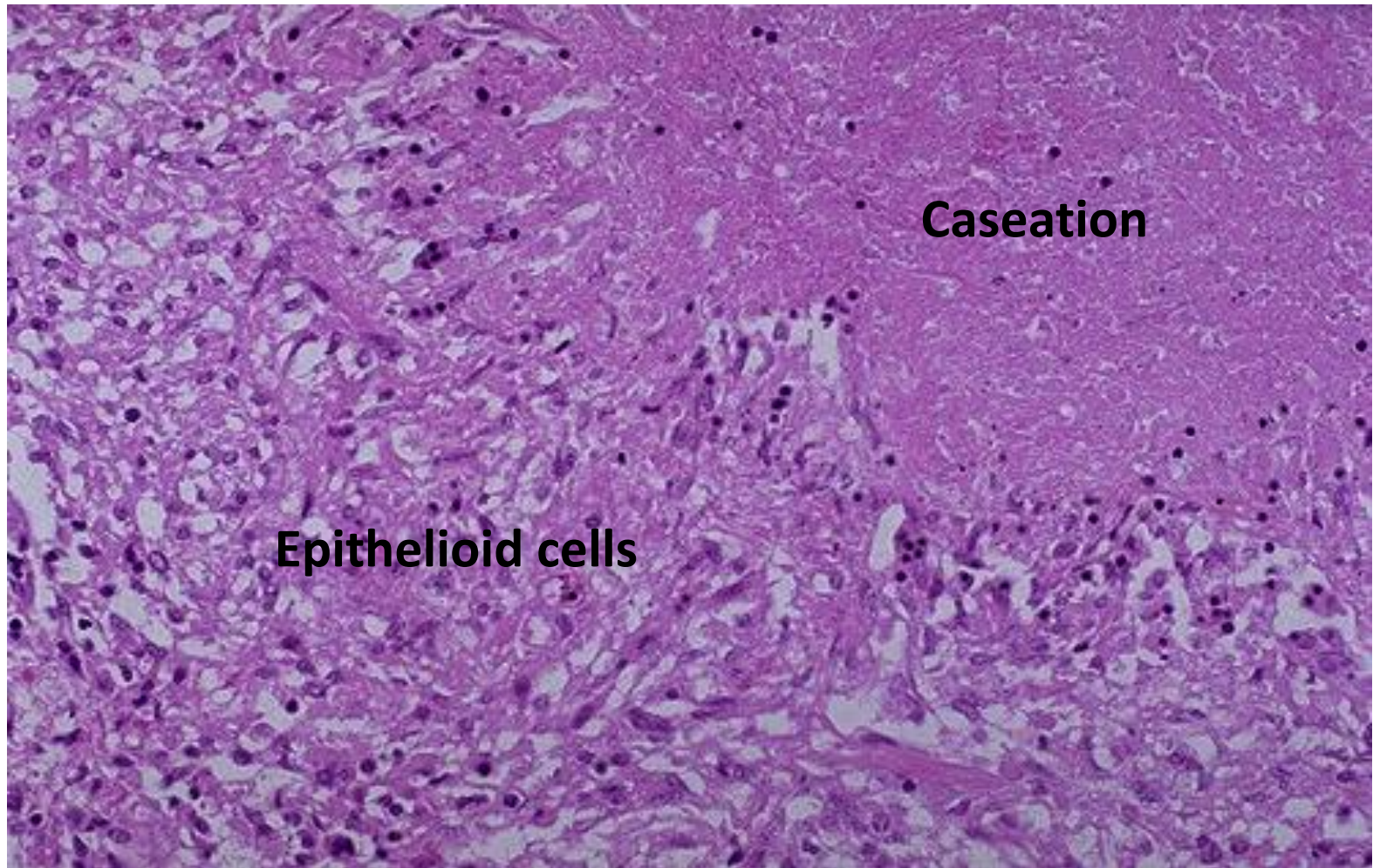
Seen under polarized light are numerous bright white crystals of talc in a patient who was an intravenous drug user.

2. Immune granulomas

- These are caused by insoluble, poorly degradable particles, typically **microbes** that are capable of **inducing a cell-mediated immune response**.
- In these responses, macrophages engulf the inciting agent, process it, and present some of it to appropriate T lymphocytes, causing them to become activated.
- The responding T cells produce:
 1. **Cytokines (IL-2)**: activates other T cells, perpetuating the response.
 2. **IFN- γ** : important in **activating macrophages** and transforming them into **epithelioid cells** and multinucleate giant cells.

- The typical example of an **immune granuloma** is that caused M. tuberculosis.
 - In tuberculosis, the granulomatous reaction is referred to as a tubercle.
 - The tubercle is classically characterized by the presence of central caseous necrosis.
 - It is necessary to identify the specific etiologic agent by:
 1. special stains for organisms (e.g., acid-fast stains for tubercle bacilli),
 2. culture methods (e.g., in tuberculosis and fungal diseases),
 3. molecular techniques (e.g., the PCR in tuberculosis),
 4. serologic studies (e.g., in syphilis).
- In sarcoidosis, the etiologic agent is unknown and the diagnosis is that of exclusion.

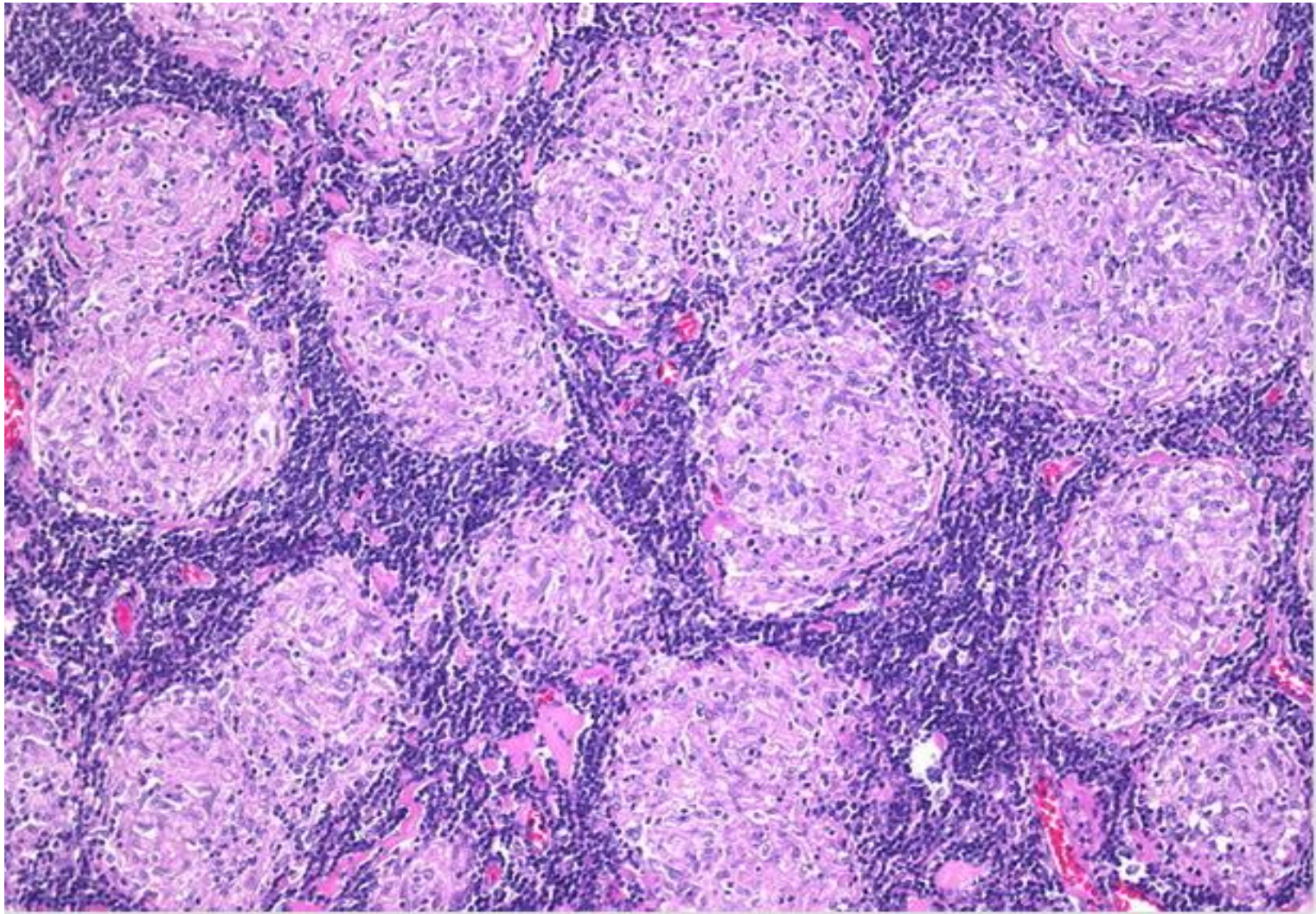
TB granulomas



Epithelioid cells

Caseation

This is a high power view showing a portion of typical TB granuloma. Note the amorphous, pinkish central caseation, which is surrounded by a rim of epithelioid cells.



Intermediate-power view shows numerous sarcoid granulomas composed of epithelioid histiocytes, occasional Langhans' giant cells and lymphocytes.

Table 3.9 Examples of Diseases With Granulomatous Inflammation

Disease	Cause	Tissue Reaction
Tuberculosis	<i>Mycobacterium tuberculosis</i>	Caseating granuloma (tubercle): focus of activated macrophages (epithelioid cells), rimmed by fibroblasts, lymphocytes, histiocytes, occasional Langhans giant cells; central necrosis with amorphous granular debris; acid-fast bacilli
Leprosy	<i>Mycobacterium leprae</i>	Acid-fast bacilli in macrophages; noncaseating granulomas
Syphilis	<i>Treponema pallidum</i>	Gumma: microscopic to grossly visible lesion, enclosing wall of macrophages; plasma cell infiltrate; central cells are necrotic without loss of cellular outline; organisms difficult to identify in tissue
Cat-scratch disease	Gram-negative bacillus	Rounded or stellate granuloma containing central granular debris and recognizable neutrophils; giant cells uncommon
Sarcoidosis	Unknown etiology	Noncaseating granulomas with abundant activated macrophages
Crohn disease (Inflammatory bowel disease)	Immune reaction against undefined gut microbes and, possibly, self antigens	Occasional noncaseating granulomas in the wall of the intestine, with dense chronic inflammatory infiltrate

THANK YOU